

THE INFANT IMPRINTING HYPOTHESIS

Commensal-Mediated Antibody Repertoire Formation in Infancy as a Determinant of Species-Specific Dietary Reactivity in Adulthood

Martin Hamer, RN (Ret.)

Hamer Glycan Systems · SGIC Whitepaper Series · August 2026

Confidentiality Notice: This document forms part of the SGIC Whitepaper Series, authored by Martin Hamer, RN (Ret.), Hamer Glycan Systems, 2026. All content is the original intellectual property of the author and is protected under Canadian copyright law. Document number to be assigned on integration with the master index.

Abstract

Anti-Neu5Gc antibodies are present in essentially all humans but differ markedly between individuals in both quantity and epitope specificity. The origin of that individual variation has not been explained. This paper proposes that it is established in infancy, during a narrow developmental window, by the interaction of two variables: the mammalian species whose glycans were present in the infant diet at weaning, and the commensal bacteria then colonising the infant nasopharynx.

The proposal rests on an established finding that dietary Neu5Gc alone is insufficient to generate these antibodies, and that commensal organisms which scavenge and display the dietary sugar are required for their induction. If the immunising presentation is bacterial, then what is imprinted is not Neu5Gc as such but the particular extended glycan configurations available at the time of colonisation — which are species-specific and therefore geographically and culturally patterned.

The hypothesis predicts that adult species-specific dietary reactivity should track childhood exposure rather than adult diet; that cow-milk formula constitutes a high-dose bovine exposure at the critical window; and that populations weaned on caprine or ovine dairy should display correspondingly different epitope repertoires with relative bovine tolerance. Explicit falsification conditions and a low-cost registry instrument for testing them are set out.

Keywords: Neu5Gc; anti-Neu5Gc antibodies; epitope repertoire; infant weaning; nasopharyngeal commensals; Haemophilus influenzae; infant formula; species specificity; SGIC; Gate One

1. The Problem This Paper Addresses

Humans cannot synthesise N-glycolylneuraminic acid (Neu5Gc) owing to an inactivating mutation in CMAH fixed in the hominin lineage.¹ Neu5Gc is nonetheless taken up from mammalian food sources and metabolically incorporated into human tissues, making it the first identified example of a xeno-autoantigen.²

Essentially all humans carry circulating antibodies against it. Critically, those antibodies are not uniform. They are polyclonal, of varying isotype and abundance, and directed against a diverse range of Neu5Gc-bearing epitopes, with the range of reactivity varying greatly between normal individuals.³

The foundational specificity paper makes a point that has been underused: the monosaccharide cannot by itself fill the binding site of an antibody, and can be presented in various linkages upon diverse underlying glycans.³ The antigen is therefore an extended structure, not a sugar. Two foods may contain identical quantities of Neu5Gc and present entirely different targets.

This creates a question that the SGIC framework has previously assumed rather than answered. Gate One of the three-gate susceptibility model requires that an individual possess antibodies recognising the particular structures present in a particular food. That framework has never explained why one person's repertoire differs from another's, nor why an individual may react reproducibly to bovine sources while tolerating caprine, ovine and porcine sources containing the same sugar.

ESTABLISHED — CMAH loss, dietary incorporation, universal presence of anti-Neu5Gc antibodies, and wide interindividual variation in epitope specificity are all published, replicated findings.

2. The Established Mechanism of Antibody Induction

The mechanism by which these antibodies arise was determined in 2010 and is not widely appreciated outside the field.⁴

Four findings from that work are load-bearing for this paper:

- 2.1** Human anti-Neu5Gc antibodies appear during infancy and correlate with weaning and exposure to dietary Neu5Gc.
- 2.2** Dietary Neu5Gc alone does not elicit anti-Neu5Gc antibodies in mice carrying the human-like deficiency. Ingestion is necessary but not sufficient.
- 2.3** No microorganism synthesises Neu5Gc. However, nontypeable *Haemophilus influenzae* (NTHi), a human-restricted nasopharyngeal commensal and opportunistic pathogen, scavenges environmental sialic acids and incorporates them into its surface lipooligosaccharide. Trace exogenous Neu5Gc — including Neu5Gc obtained from commercial baby food — is taken up and displayed by this organism.
- 2.4** NTHi displaying Neu5Gc-containing lipooligosaccharide induces anti-Neu5Gc antibodies in Neu5Gc-deficient mice without added adjuvant. In infants, anti-Neu5Gc antibodies appear coincident with antibodies against NTHi.

A subsequent finding gives an indication of the strength of the resulting response. NTHi has evolved preferential utilisation of the human sialic acid: its CMP-sialic acid synthetase shows approximately four thousand-fold higher catalytic efficiency for Neu5Ac than for Neu5Gc, apparently avoiding display of the non-human form. The proposed selective pressure is the human antibody response itself.⁵

ESTABLISHED — The bacterial cooperation model of antibody induction, and the subsequent demonstration of bacterial adaptation away from Neu5Gc display, are published in peer-reviewed immunology and microbiology literature. Neither is the author's work.

3. The Hypothesis

If the immunising event is bacterial display of scavenged dietary Neu5Gc during infancy, then what the developing immune system learns to recognise is not Neu5Gc in isolation. It is the set of extended glycan configurations available for scavenging at that moment, presented on a bacterial surface in an inflammatory context.

Those configurations are species-specific. Bovine, caprine, ovine and porcine glycoconjugates differ in linkage, in underlying glycan structure, and in scaffold class. Bovine milk delivers Neu5Gc predominantly on milk fat globule membrane gangliosides; bovine muscle delivers it predominantly on glycoprotein N-glycans. These are different antigens presenting the same monosaccharide.

The hypothesis is therefore stated as follows:

Adult species-specific reactivity to mammalian dietary antigens is substantially determined by the mammalian species whose glycans were present during the infant weaning window, imprinted via commensal display, rather than by adult dietary exposure.

Three corollaries follow.

- 3.1** Reactivity patterns should cluster by the geography and culture of early childhood rather than by current residence or current diet, because dairying species differ regionally and traditional weaning foods follow them.
- 3.2** Cow-milk infant formula constitutes an unusually high, sustained and early bovine glycan exposure delivered precisely into the imprinting window. Human milk contains no Neu5Gc; the difference between exclusive breastfeeding and cow-milk formula is therefore not a difference of degree but of antigenic kind.
- 3.3** The effect should be strongest in birth cohorts predating the globalisation and standardisation of commercial weaning foods, and may be attenuating in younger cohorts.

HYPOTHESIS — The imprinting proposal is the author's and is untested. The mechanism it extends is established; the extension is not.

4. Supporting Observations Already in the Literature

Three published findings sit adjacent to this hypothesis without having tested it.

First, a xenotransplantation cohort of eighty-four healthy individuals found variability in anti-Neu5Gc antibody levels associated with prior vaccination history and with the geographic region in which the individual spent childhood.⁶ The study measured total levels rather than epitope specificity, and did not examine infant feeding. That childhood region rather than current residence emerged as a variable is nonetheless notable, since the study had no theoretical reason to expect it.

Second, a large French dietary cohort demonstrated that Neu5Gc intake, sex and age affect the specificity, levels and repertoires of anti-Neu5Gc IgG, though not their affinity, with measurable

differences in α 2-3 versus α 2-6 linkage recognition between lowest and highest consumers.⁷ Diet demonstrably shapes which structures are recognised. That study examined adult intake.

Third, sialoglycan microarray methodology capable of resolving specificity against dozens of defined structures simultaneously is established, standardised and in routine use.⁸ The instrument required to test this hypothesis already exists and requires no development.

The combination — species-resolved epitope repertoire against documented infant feeding history and childhood geography — does not appear to have been examined.

5. Predictions

The hypothesis generates the following testable predictions, stated so that each can fail.

#	Prediction	Test
P1	Species-specific epitope binding clusters by childhood geography, not adult residence or adult diet.	Array + childhood history
P2	Cow-milk-formula-fed cohorts show bovine-dominant repertoires relative to exclusively breastfed cohorts with delayed mammalian introduction.	Array + feeding history
P3	Individuals weaned on caprine or ovine dairy show caprine/ovine epitope binding with relative bovine tolerance — the inverse of the Northern European pattern.	Array, Mediterranean cohort
P4	The association is stronger in pre-1980 birth cohorts than in later ones.	Stratify by birth decade
P5	Reported food reactions correspond to the array-defined repertoire rather than to measured Neu5Gc content of the foods.	Registry + array

P5 is the prediction that bears most directly on the wider SGIC framework, because the author's own longitudinal record shows a severity ordering that runs opposite to the published Neu5Gc content ordering: beef is highest in content and produces the mildest response, while concentrated dairy derivatives lower in content produce the most severe.

6. Proposed Test One: Registry Instrument

The SGIC Patient Outcomes Registry already collects self-reported dietary trigger patterns. Four additional intake questions would render this hypothesis testable at negligible cost:

Q1 Country and region of birth.

Q2 Decade of birth.

Q3 Infant feeding: exclusively breastfed, exclusively formula-fed, mixed, or unknown. If formula, brand or base if known.

- Q4** Which animal's milk was routinely present in the household during early childhood (cow, goat, sheep, buffalo, camel, mare, none, unknown).

No existing dataset pairs species-specific reaction reports with infant feeding history. Collecting these four fields would make the registry the only instrument capable of examining P1 through P4 at population scale, and would do so prospectively rather than by retrospective chart review.

7. Proposed Test Two: Stratified Sialoglycan Array

The definitive test requires serum and an array. The design is array-first, consistent with the principle adopted across this programme: the antibody profile is measured and the prediction recorded before any dietary claim is evaluated.

- 7.1** Recruit participants stratified by childhood exposure group: cow-dairy region, sheep/goat-dairy region, low-dairy region; formula-fed versus exclusively breastfed where ascertainable.
- 7.2** Profile serum IgG on a sialoglycan microarray including bovine-type, caprine-type, ovine-type and porcine-type Neu5Gc-bearing structures with matched Neu5Ac controls, including ganglioside headgroups and glycoprotein N-glycan structures separately.
- 7.3** Record predicted reactive species per participant before collecting dietary reaction history.
- 7.4** Compare predicted against reported reactivity.

The author's own serum constitutes an immediate single-subject test. The prediction is preferential binding to bovine-type structures, with binding to caprine, ovine and porcine structures at or near control levels.

8. Falsification Conditions

The following outcomes would falsify or substantially damage the hypothesis and are stated in advance.

- F1** Uniform epitope binding across childhood-exposure groups would falsify the imprinting model outright.
- F2** If adult dietary intake predicts epitope repertoire better than infant exposure does, the hypothesis is reduced at best to a minor contributor.
- F3** If the author's own serum binds caprine, ovine and porcine structures as strongly as bovine, then his twenty-three-year species-specific observation has some other explanation, and this paper is wrong in the case that generated it.
- F4** If antibody repertoires prove equally diverse in populations with negligible mammalian dietary exposure, bacterial sialidase scavenging from non-dietary sources would be the better explanation and dietary imprinting would be redundant.

9. Limitations

Commensal organisms other than NTHi possess sialidases capable of metabolising Neu5Gc, which may supply the antigen even where dietary intake is low, blurring any geographic signal.

Infant feeding history is subject to recall bias, particularly in older participants reporting on their own infancy at second hand. Where possible this should be recorded as reported and flagged as such rather than treated as documented fact.

A systematic review has identified conflicting prevalence estimates for these antibodies and an absence of standardised quantification methodology, which complicates comparison across cohorts and laboratories. Any multi-site work under this hypothesis should use a single array platform and a single laboratory.

Childhood geography is confounded with genetic ancestry, HLA type, and a great many other exposures. Formula feeding is confounded with socioeconomic status, maternal health and birth cohort. Neither association, if found, would establish causation.

Commercial formula composition has changed substantially over the decades, including the recent deliberate addition of bovine milk fat globule membrane to some preparations. Cohort effects should be expected and may be large.

10. Relation to the Existing Framework

This paper supplies a mechanism for Gate One of the three-gate susceptibility model, which has previously been stated as a requirement without an account of its origin. Under the imprinting hypothesis, Gate One is set in the first two years of life by variables over which the individual has no control and of which they retain no memory.

It also reframes the programme as a whole. The framework has been presented as an account of what people eat. If this hypothesis holds, it is more accurately an account of what a person's immune system learned to recognise before they could speak, and of what they have gone on eating ever since.

No revision to the epitope/configuration correction, the MFGM transport correction, or the three-gate model is required. This paper extends them; it does not amend them.

References

1. Chou HH, Takematsu H, Diaz S, et al. A mutation in human CMP-sialic acid hydroxylase occurred after the Homo–Pan divergence. *Proc Natl Acad Sci USA*. 1998;95(20):11751–11756.
2. Tangvoranuntakul P, Gagneux P, Diaz S, et al. Human uptake and incorporation of an immunogenic nonhuman dietary sialic acid. *Proc Natl Acad Sci USA*. 2003;100(21):12045–12050.
3. Padler-Karavani V, Yu H, Cao H, et al. Diversity in specificity, abundance and composition of anti-Neu5Gc antibodies in normal humans: potential implications for disease. *Glycobiology*. 2008;18(10):818–830.
4. Taylor RE, Gregg CJ, Padler-Karavani V, et al. Novel mechanism for the generation of human xeno-autoantibodies against the nonhuman sialic acid N-glycolylneuraminic acid. *J Exp Med*. 2010;207(8):1637–1646.
5. Nontypeable *Haemophilus influenzae* has evolved preferential use of N-acetylneuraminic acid as a host adaptation. *mBio*. 2019;10(3):e00422-19.
6. Zhang Z, Hara H, Long C, et al. Anti-Neu5Gc and anti-non-Neu5Gc antibodies in healthy humans. *PLoS One*. 2017;12(7):e0180768.

7. Bashir S, Fezeu LK, Leviatan Ben-Arye S, et al. Association between Neu5Gc carbohydrate and serum antibodies against it provides the molecular link to cancer: French NutriNet-Santé study. *BMC Med.* 2020;18(1):262.
8. Leviatan Ben-Arye S, Yu H, Chen X, Padler-Karavani V. Profiling anti-Neu5Gc IgG in human sera with a sialoglycan microarray assay. *J Vis Exp.* 2017;(125):56094.
9. Zaramela LS, Martino C, Alisson-Silva F, et al. Gut bacteria responding to dietary change encode sialidases that exhibit preference for red meat-associated carbohydrates. *Nat Microbiol.* 2019;4(12):2082–2089.
10. Altman MO, Gagneux P. Absence of Neu5Gc and presence of anti-Neu5Gc antibodies in humans — an evolutionary perspective. *Front Immunol.* 2019;10:789.
11. A systematic review reveals conflicting evidence for the prevalence of antibodies against the sialic acid xenoautoantigen Neu5Gc in humans and the need for a standardised approach to quantification. *Front Mol Biosci.* 2024;11:1390711.

Reference details require verification against source before publication.